Research Policy 47 (2018) 1060–1069

Contents lists available at ScienceDirect

Research Policy

journal homepage: www.elsevier.com/locate/respol

University licensing and the flow of scientific knowledge☆

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ARTICLE INFO

JEL classification:
O31
O32
R12

Keywords:
University research
Technology transfer
Scientific communication
Intellectual property
Licensing
Material transfer agreements
Research inputs
Research tools

ABSTRACT

As university involvement in technology transfer and entrepreneurship has increased, concerns over the patenting and licensing of scientific discoveries have grown. This paper examines the effect that the licensing of academic patents has on journal citations to academic publications covering the same scientific research. We analyze data on invention disclosures, patents, and licenses from the University of California, a leading U.S. academic patenter and licensor, between 1997 and 2007. We also develop a novel "inventor-based" maximum-likelihood matching technique to automate and generalize Murray's (2002) "patent-paper pairs" methodology. We use this methodology to identify the scientific publications associated with University of California patents and licenses.

Based on a "difference-in-differences" analysis, we find that within our sample of patented academic discoveries, citations to licensed patent-linked publications are higher in the three years after the license, although this difference is not statistically significant. We then disaggregate our sample into (a) patented discoveries that are likely to be used as "research tools" by other researchers (based on the presence of material transfer agreements (MTAs) that cover them) and (b) patented discoveries not covered by MTAs. Citations to publications linked to licensed patents in the latter subset (not covered by MTAs) are higher for publications linked to licensed patents, and this difference is statistically significant. In contrast, licensing of patented discoveries that are also research tools is associated with a reduction in citations to papers linked to these research advances, raising the possibility that licensing may restrict the flow of inputs to "follow-on" scientific research.

1. Introduction

Growth in patenting of academic research advances in U.S. and other universities within the OECD has triggered considerable debate since at least 1980, the year of passage of the U.S. Bayh-Dole Act. Supporters contend that patenting and licensing of university inventions can speed the transfer of scientific discoveries to the private sector, promoting the commercialization of such advances. On the other hand, critics predict a collision between the norms of science and the norms of commerce, arguing that the exclusionary effects of patents will slow the progress of science. Despite the importance of this question, there has been relatively little empirical work on the extent to which the patenting of academic research results affects scientific research progress. The modest volume of this work (e.g., Fehder et al., 2014; Murray and Stern, 2007) reflects the challenge of obtaining data that can be used to examine any such effects.

This paper develops a new approach to matching scientific publications and patents, and employs this methodology to examine the effect of

☆ We are grateful to Wendy Streitz, Ron Franck, and other employees of the UC Office of the President for access to the intellectual property data employed in this paper, and to Thompson Reuters for its data on scientific publications. We also thank Fiona Murray and Scott Stern for sharing their data. This work has benefited greatly from comments and suggestions by Brian Wright, Bronwyn Hall, Christian Helmers, Fiona Murray, Guido Bünstorf, Heidi Williams, Janet Bercovitz, Jennifer Kuan, Jian Wang, Joanne Oxley, Martin Kenney, Michael Frakes, Paul David, Rebecca Eisenberg, Rob Seamans, Robin Rasor, Rosemarie Ziedonis, and two anonymous reviewers and participants in seminars at the Copenhagen Business School, UC Berkeley, North Carolina State University, the Ohio State University, Santa Clara University, the University of South Carolina, Stanford University, the University at Buffalo, the University of Rhode Island, the 2014 USPTO-Searle Center Conference on Innovation Economics at Northwestern University, 2011 NBER Economics of Intellectual Property Policy and Innovation Workshop, the 2011 Georgia Tech Roundtable for Engineering Entrepreneurship Research, the 2012 Academy of Management annual meeting, the 2014 Economics of Knowledge Contribution and Distribution Workshop, the 3rd ZEW Conference on the Dynamics of Entrepreneurship, the Universität Bocconi Workshop on Market and Non-Market Mechanisms for the Exchange and Diffusion of Innovation, the 2016 Sumantra Ghoshal Strategy Conference, and the 2016 Munich Summer Institute. Robert Picard provided expert programming assistance. The authors gratefully acknowledge financial support from the Institute for Business Innovation at the Haas School of Business, the National Science Foundation, the Industrial Partnerships Office at Lawrence Livermore National Laboratory, the Program in Law, Economics, and Technology at the University of Michigan Law School, the Rotman School of Management at the University of Toronto, and the Sloan Foundation. We also appreciate the support of the Stanford Institute of Economic Policy Research and Harvard Business School.

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https://doi.org/10.1016/j.respol.2018.03.008
Received 7 February 2017; Received in revised form 12 March 2018; Accepted 12 March 2018
Available online 25 April 2018
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licensing on the journal citations to related publications. We interpret increased citations to these scientific publications after a license issues as evidence that licensing is correlated with greater prominence and use by other scientists of the knowledge embodied in the paper, while decreases in such citations may indicate a restrictive effect of licensing on scientific communication. Drawing on related work by Walsh et al. (2007), we also examine the effect of licenses on related scientific publications covering inputs to the experiments of other researchers, a class of research results often referred to as "research tools."

The next section of this paper discusses the use and potential effects on scientific research of formal intellectual property rights covering academic discoveries. We describe our data and methodology in Section 3. We present the results of our analysis in Section 4. Finally, we discuss the implications of our results and conclude with a summary of the contributions and limitations of this paper in Section 5.

2. Research and intellectual property in academia

Universities have long been important performers of research, particularly basic research. U.S. universities have expanded their performance of basic research since the 1970s, performing almost 49% of U.S. basic research in 1979 (National Science Foundation, 2012, Table 4–4) and almost 59% in 2006 (National Science Board, 2016, Appendix Tables 4–7, pp. 4-150–4-156). Expansion in their role in basic research has coincided with increased patenting and licensing of U.S. university discoveries, particularly in biotechnology.

The roles of universities as sources of both basic knowledge and potentially valuable ingredients for commercial innovation raises the possibility of conflict between these roles, with detrimental consequences for the advance of scientific research. Do patents and licenses restrict access to such knowledge? Or is the existence of a patent and/or a license for that patent a signal of the quality or commercial potential of scientific work that leads to greater exploration of the area?

Our examination of the effects of patenting and licensing on scientific communication focuses on "patent-linked publications" (Murray, 2002). These are discoveries that are published in scientific journals and become the subject of successful patent applications.

2.1. The effects of intellectual property protection and licensing on scientific research and communication

The growth of academic patenting has been the subject of a large literature and considerable debate over its effects on the scientific research enterprise (e.g., Mowery et al., 2004). Heller and Eisenberg (1998) argue that expanded patenting of academic research results may result in fragmented and overlapping property rights covering biomedical research, limiting the ability of scientists to access patented and licensed research outputs for follow-on research. Other scholars raising concerns over the expanded assertion of property rights in science include Nelson (2004) and David (2004). Empirical research seeking to assess the effects of patenting on scientific communication has examined the effects of patenting on biomedical researchers' willingness to share information on their work (Blumenthal et al., 1997; Campbell et al., 2002). More recent research has analyzed the effects of patenting biomedical discoveries that are also disclosed in scientific papers. Some of this work finds that the issuance of a patent resulted in modest but significant declines in citations to the research papers related to the patent (Murray and Stern, 2007; Sampat, 2005).[1] Similarly, Williams (2013) finds that patenting of genes by the private firm Celera reduced citations to related scientific research. Other work, however, argues that academic researchers rarely if ever search to determine whether a prospective research project or experiment will infringe on patents (Lei et al., 2009; Walsh et al., 2007), raising questions about the mechanism behind any observed citation decreases.

As the previous paragraph indicates, the majority of prior empirical research on the effects on scientific communication of academic patenting and licensing of discoveries has focused mainly on patenting of academic research results. The effects of university licensing of these patents have received much less attention from scholars.

Why and how might licenses for a specific research advance affect the behavior of other researchers not involved as authors or patent holders? Sampat and Ziedonis (2004) examined patent citations to Columbia University and University of California patents that were licensed and found that citations were higher for licensed than for unlicensed patents.[2] These scholars interpreted this pattern of increased patent citations as reflecting growth in patenting (and potentially, research) activity in the technological area(s) surrounding the licensed patents. More recently, Drivas et al. (2017) found that non-licensees increased their citations to patents exclusively licensed (based on restrictions covering either geographic area or field of use) by the University of California after the licenses were executed. Similar to Sampat and Ziedonis (2004), Drivas et al. (2017) regard the increase in non-licensee patent citations as a reaction to the potential commercial value signaled by the negotiation of the license.[3]

A similar signaling effect associated with the execution of a license could increase citations to publications linked to a patented and licensed discovery. In such a case, the issue of a license would "demonstrate" that a particular area of research has scientific value, leading other investigators to pursue work in related fields. It is also possible (see Larsen, 2011) that contemporary academic researchers may choose research areas partly based on their potential commercial value, and therefore might respond positively to a "signal" that a given area of research has attracted the attention of industrial licensees. Regardless of whether any such signal operates through perceptions among researchers of scientific or commercial potential, this argument predicts an increase in citations to publications linked to patented disclosures following the negotiation of the license.

Equally plausible arguments, however, suggest a chilling effect of licensing on scientific communication. Reactions by university technology licensing offices and/or their licensees to any evidence of patent infringement (even for research purposes, inasmuch as the legal status of any "research exemption" from such infringement suits remains unclear) could be swifter and stronger in the case of patents that are licensed. And licenses may include provisions for reach-through royalties and limitations on the management of intellectual property resulting from follow-on research. Moreover, the negotiation of a license may take considerable time, delaying other researchers' access to the materials or tools embodied in the disclosure.

We are thus agnostic on the likely direction of any effect of licenses on scientific communication associated with publications linked to licensed academic patents. Indeed, both effects may be present for papers in various fields of research, and our findings could shed light on the magnitude of any offsetting effects.

[1] In follow-on work, Fehder et al. (2014) find that the negative effect is most pronounced for patent-related papers published in the early years of a journal's founding, leading them to conclude that this effect is greatest before a journal has established a reputation for publishing "high quality" science.

[2] Although Sampat and Ziedonis, Sampat and Ziedonis do not differentiate between pre- and post-license citations in their regressions, they do report that 86% of citations to licensed patents in their sample occur after the license.

[3] Ziedonis (2007) reports that licenses negotiated by the University of California are confidential. Firms must sign a "non-disclosure" agreement prohibiting the disclosure of the license or use of the University's name in marketing of products arising from the licensed discovery. In exchange, the University agrees to not disclose the existence of the license or the identity of the licensee. Under such confidentiality provisions, one would expect a "signal" to be weak or non-existent. According to our conversations with a director of a leading public research university technology transfer office, however, despite such secrecy provisions in licensing, the existence of and identity of licensees are typically widely known through informal conversations among academic scientists, "freedom of information act" requests for licensing data from public universities, and other means.

2.2. The importance of access to research inputs

Patents and licenses increase the "excludability" of intellectual property, exposing other researchers to potential legal liability in the event that they utilize or exploit the intellectual property protected by patents issued to others for their own research or commercial use. A different form of excludability, highlighted by Walsh et al. (2007), occurs when one researcher denies physical access to materials or other research results that are inputs to the experiments of another researcher. The survey results of Walsh et al. (2007) indicate that such denials can impose significant costs and delays on the scientific work of other researchers, costs and delays that according to these authors, exceed those associated with patents.

One class of research inputs is widely (and imprecisely) identified as "research tools." For example, the NIH Working Group on Research Tools (1998) defines them as "the full range of resources that scientists use in the laboratory … the term may thus include cell lines, monoclonal antibodies, reagents, animal models, growth factors, combinatorial chemistry libraries, drugs and drug targets, clones and cloning tools (such as PCR), methods, laboratory equipment and machines, databases and computer software."[4]

As we noted above, Walsh et al. (2007) argue that denials of access to such research tools are more likely when the erstwhile supplier of them is engaged in "business activity," such as licensing of the invention disclosure associated with the tools.[5] Interviews with researchers conducted for this study support this view, with one academic scientist involved in a start-up firm saying "if another company asked to use our [materials] for [the same purpose as our company uses them] we would say 'no' ".[6]

Even when permission is granted, however, researchers gaining access to research tools that are associated with licensed disclosures may encounter other difficulties, as highlighted by the celebrated case of the Oncomouse (Murray, 2010):

> "In 1984, scientists at Harvard University carefully engineered a new mouse to have a predisposition to cancer, the Oncomouse …The Harvard researchers …patented their creation and subsequently licensed this patent to DuPont …
> [Dupont] set a high price per mouse …placed restrictions on breeding programs …demanded publication oversight …[and] insisted upon a share of any commercial breakthroughs made using the Oncomouse."

These and similar restrictions may limit the use by other researchers of research tools such as the Oncomouse. Moreover, as the NIH Working Group on Research Tools (1998) noted in its report, licensees may have an incentive to restrict access to these materials:

> "If the sponsor or licensee plans to develop the research tool as a commercial product for sale to researchers, it may be unwilling to permit the university to undercut its position in this particular market by giving the tool away to potential paying customers."

Surveys of scientists by Walsh et al. (2007) and Lei et al. (2009) find that requests by researchers for research tools from industrial researchers, a group more likely to be engaged in commercial applications of research, were rejected approximately twice as often as requests to other academics.

As this discussion of "research tools" indicates, developing a definition that facilitates their identification and empirical analysis is challenging. The definitions of research tools employed by the NIH Working Group and Walsh et al. (2007) are very broad and do not lend themselves to empirical operationalization using administrative data. Instead of attempting to develop and defend a definition of research tools that relies on the characteristics of the relevant invention disclosure or patent, we identify research tools based on the existence of a Material Transfer Agreement (MTA) associated with a patented invention disclosure.

Material Transfer Agreements (MTAs) are agreements that govern the transfer and exchange of materials used in research. Although the informal exchange by researchers of biological materials for use in fundamental research has a long and occasionally controversial history in the biomedical sciences, these materials exchanges historically were governed by little more than a letter from the source accompanying the materials, requesting acknowledgement and in some cases asking that the materials not be passed on to third parties (McCain, 1991). The more elaborate MTAs used in many contemporary materials exchanges appear to be a byproduct of the post-1980 surge in academic patenting (Streitz and Bennett, 2003).

It is important to highlight the differences between MTAs and formal licenses. The "research tools" that typically are covered by MTAs are inputs to other research activities, rather than a discovery that may have commercial potential. Research tools frequently are linked to licensed research advances, but both the intent and the content of licenses and MTAs differ from one another. Licensing of academic research advances generally is limited to patented discoveries, but many MTAs cover unpatented research results, including cell lines, data, and physical artifacts. Many if not most licensing agreements contain extensive language covering the financial terms under which the user of a patented academic discovery is legally bound to compensate the licensor institution, while MTAs rarely include royalty provisions beyond occasional "reach-through" provisions covering the use by recipient researchers of a research input in a research advance that is itself licensed for royalties. Academic research projects may generate artifacts that are transferred to other researchers under the terms of an MTA, but produce nothing that is patented and/or licensed.

We use the presence of an MTA to proxy whether an invention disclosure is a research tool.[7] To confirm that MTAs are a good indicator that an invention disclosure is associated with a research tool, we examine UC Berkeley data on incoming MTAs—that is, the agreements associated with research materials requested from other researchers by UC Berkeley researchers during 2004–2008. These data describe the requested materials and their intended use by the recipient UC Berkeley researcher. We analyzed a random sample of 50 of these MTAs, and found that 44% were related to DNA/RNA/Plasmids, 32% concerned cell lines or other biological/chemical agents, 16% were animal models, 6% were data transfers, and 2% were concerned with "other non-research inputs." Overall, therefore, 98% of these MTAs

[4] The National Research Council panel has a similarly broad definition that includes materials that "…may be critical inputs for the success of a research project." (National Research Council, 2010; .

[5] Walsh et al. (2007) define an academic scientist receiving a request for research inputs as being involved in business activity when he or she "had been involved in negotiations over rights to their inventions; had begun developing a business plan or laying other groundwork for starting a firm; had a startup based on their invention; had a product of process in the market; *or had some licensing income*" (p. 1196) (italics added).

[6] Even though this scientist was unwilling to share materials in this instance, in other instances he/she had shared materials.

[7] It is important to emphasize that we use MTAs to identify a research tool but do not attempt to more precisely capture the "effect" of an MTA on the communication of scientific research results. Among other things, such an analysis would require a more comprehensive sample of MTA-linked disclosures, including those for patented and unpatented disclosures, than we have. One of the few analyses of the role of MTAs in the scientific research enterprise is Scott Stern's discussion of biological resource centers (Stern, 2004). Biological resource centers (BRCs) are nonprofit materials depositories that play a key role in maintaining the reliability and provenance of cell lines used by industrial and academic researchers-as Stern notes, contamination of widely used cell lines has caused major research fiascoes in the past several decades. Stern argues that the use of MTAs by BRCs has aided the exchange of materials, and recommends that MTAs be a standard complement to patents covering biological discoveries: "Putting MTAs in place at the time of patent approval lowers the cost of mutually beneficial transactions between the developers of materials and follow-on researchers and widens the availability of patented biomaterials" (Stern, 2004; pp. 96–97). Stern thus argues that MTAs can facilitate the integrity and exchange or research materials, enhancing the efficiency of the research enterprise.

involved materials that fit within the NIH Research Tools Working Group definition of research tools. We also analyzed the descriptions of the intended use for the materials requested through the MTA by UC Berkeley researchers, and found that 94% of the MTAs indicated that the requested material was to be used as an input to further research and a further 4% implicitly indicated that such uses were intended.

This analysis of UC Berkeley MTAs leads us to conclude that our treatment of the presence of an MTA as an empirical indicator that a given invention disclosure is a research tool and/or input to follow-on scientific research is defensible. It is important nevertheless to note two caveats associated with our empirical use of MTAs as indicators of research tools. As Walsh et al. (2007) and Mowery and Ziedonis (2007) point out in their discussions of MTAs, a majority of the materials transfers among academic scientists do not rely on formal MTAs that are disclosed to academic Technology Transfer Offices (TTOs). Although we believe that the presence of an MTA is a reasonable indicator that a given disclosure has applications as a research tool, our data in fact contain many other disclosures (including disclosures that are patented and licensed) that may well be research tools but (lacking an MTA) cannot be identified as such. It is likely that our empirical approach thus understates the effects on citations of licensing of patented disclosures that are research tools. In addition, our data enable us to only identify the effects of those MTAs that are negotiated and agreed to by all parties to the materials transfer. In other words, and in contrast to Walsh et al. (2007), we do not identify the effects on scientific research of the denial by researchers of other researchers' requests for research materials, nor do we attempt to explain why MTAs might affect the dissemination of these scientific results within the research community. Instead, we are solely concerned with analyzing the effect of licensing on research tools.

Our empirical research strategy thus addresses two questions: (1) Using citations to related scientific papers as a proxy for scientific communication, do licenses on patents tend to increase or decrease such communication, and (2) how does negotiation of a license for a disclosure linked to a research tool affect citations to related research papers?

3. Data and methodology

This section describes our data and methodology. We draw on two principal sources of data for our empirical analysis. The first, the "IP data," is an extract from the technology disclosure database maintained by the technology transfer office within the University of California Office of the President (UCOP). UCOP monitors and in some cases manages invention disclosures, patent applications, and licensing transactions for nine campuses of the University of California (including five medical schools) during the period of our study.[8]

These data list 11,341 inventions reported by University of California faculty from 1997 to 2007. These invention disclosures led to 2035 issued U.S. patents, 1890 licenses to these patents, and 3853 MTAs by the end of 2009. Note that only a small subset of technology disclosures is patented, and universities' patenting propensity varies among fields of academic research—since the 1980s, patenting and licensing activity at UC has been dominated by biomedical research (Mowery et al., 2004). The distribution of MTAs also is highly skewed, with few disclosures generating the majority of MTAs and many disclosures associated with no MTAs. This echoes the finding in Mowery and Ziedonis (2007) that MTAs are disproportionately concentrated in biomedical fields of research, as are licenses.

The second source, "publications data," comes from the Web of Science, an internet-based service that tracks the bibliographic information and the citations to and from articles published in 10,000 of the highest-impact journals across 256 disciplines.[9] This database supplied the title, author names, journal, publication date, and citation information for each scientific paper. The information on "forward citations," citations from subsequent published papers to that publication, was extracted through the end of 2009. The Web of Science also provides a number of well-accepted measures of journal quality. The most prominent of these is an "impact factor" that measures the average number of times an article in that journal is cited in its first two years, and which we include in our analysis.

Observing the relationship between the licensing of a patent on citations to a related publication requires identifying the linkages among publications, patents, and licenses. The connections among disclosures, patents, and licenses are contained in the data provided by UCOP, which tracks these directly. But the UC data themselves provide no direct link between published papers and any patents covering the research advances covered by them—what Murray and Stern (2007) termed "patent paper pairs." Prior work linking patents to related journal articles has relied on expert judgment (Murray, 2002; Murray and Stern, 2007) or matched publication characteristics to scientific references in patents (Azoulay et al., 2012). We instead employ an "inventor-based matching" methodology to link patents to scientific papers. This technique assumes that inventors listed on a patent are likely to be authors of related publications. We thus construct a maximum-likelihood estimator for the publication(s) that best match a particular patent. We describe this technique in greater detail in the Appendix.[10]

3.1. Construction of control groups

Designing a specification to test the effect of licensing on scientific communication is a challenging exercise, because of the complex and varied patterns of citations that publications may receive during our sample period. Citations to some publications may grow throughout the period covered by our data, whereas for other publications, citations may rise, peak, and then decline. Moreover, the timing and rate of any ascent and decline in citations may vary. It is difficult to construct a sufficiently flexible parametric model to adequately accommodate these differences, yet it is essential to accommodate variations in citation patterns in order to construct a credible control sample and avoid spurious statistical results.[11] Because of the complex and potentially non-linear interactions of various control variables that we employ to construct treatment and matching control groups, we pursue a flexible nonparametric approach. This technique allows us to weaken the assumption of linearity and enables us to account for the many plausible interaction effects that may be present in our analysis.

The non-parametric method we use is "nearest neighbor" matching. It searches the set of non-treatment observations to identify the one "closest" to each treated observation, based on the characteristics of nontreatment and treated observations. Collectively, these "closest" non-treatment observations form a control group. Because this search is

[8] A tenth campus was established at Merced in 2005, but no invention disclosures leading to patents originated from that campus during the period covered by this study.

[9] http://thomsonreuters.com/products_services/science/science_products/a-z/web_of_science (downloaded May 2010).

[10] As we explain in detail in the Appendix, we tested our methodology using the patent-paper pairs generated by Murray and Stern (2007) through hand matching. Our methodology found identical patent-paper pairs for 95% of their sample.

[11] For example, it is reasonable to assume that publications in more highly cited journals accrue more citations, which would argue for including *Journal Impact Factor* as a control variable. Similarly, the academic discipline (hereafter *Journal Subject*) could also drive citation patterns, as could patterns of citations to a publication prior to the license (*Citations in* $t-1$, *Citations in* $t-2$). Each of these the effects could be non-linear, which suggests the inclusion of higher-order terms. Interaction terms among these variables would also be important, since the effects of *Journal Impact Factor* could differ by discipline, to cite only one example. Each of these effects could affect a publication differently at different points in time, calling for them to be interacted with age fixed effects. Fully interacting all of these effects would result in many of the parameter estimates being determined by small numbers of observations, or lacking observations at all.

done based on observable characteristics, the control and treatment observations should be similar along these dimensions.[12]

We identify these "nearest neighbors" using the "Matching" package for R software,[13] which employs the evolutionary search algorithm "genetic matching" to pair treatment observations with potential controls and iteratively improve covariate balance (Diamond and Sekhon, 2013; Sekhon, 2011).[14] According to Sekhon (2011), genetic matching "dominates the other matching methods in terms of MSE [Mean Squared Error] when assumptions required for EPBR [Equal Percent Bias Reduction] hold and, even more so, when they do not."[15]

Our procedure employs two types of variables (observable characteristics), those where we specify an exact match and those where a nearby (nearest neighbor) match is sufficient. We require an exact match on the following variables:

- Publication Age: Number of years between paper publication date and year of the observation.
- Journal Subject: Academic discipline of the journal (e.g., medicine).
- Patent Granted (yes/no): Whether the related patent has been granted at the time of license.
- MTA Issued (yes/no): Whether the paper has an associated MTA at the time of the license.

For example, these restrictions imply that in our analysis a (licensed) treatment observation in the life sciences with an issued patent and no MTA would be compared with a non-licensed control group observation that also has those characteristics (i.e., (1) it is in the life sciences, (2) it is associated with an issued patent, (3) it has no MTA, and (4) the number of years since publication is identical for the treatment and control observations).

For each treatment observation that matches on these exact characteristics, we then choose its nearest neighbor based on its proximity in the following five characteristics:

- Journal Impact Factor.
- Publication Year.
- Citations in $t-2$: Number of citations two years before the treatment.
- Citations in $t-1$: Number of citations one year before the treatment.
- Slope of citation curve between $t-2$ and $t-1$.

In each of these dimensions we limit the maximum "distance" between each treatment and control observation to one standard deviation for that variable (the "caliper"), thus excluding any observation that differs by more than that amount for any characteristic.[16] Under this procedure, treatment observations with no equivalent control observation are dropped from the sample. Collectively, these restrictions produce control observations such that for each characteristic, the control matches either exactly or within one standard deviation to the corresponding treatment observation.[17]

Before we estimate the effects of licensing on citations to patented publications, we must verify the covariate balance between the treatment and control groups. An effective matching procedure should yield summary statistics for the treatment and control groups that are similar. Table 1 reports the mean of each variable for the treatment and control groups produced by genetic matching and the results of *t*-tests (difference of means) and Kolmogorov–Smirnov (*KS*) tests (difference in distributions) between these two groups.

The means and distributions for the first four variables reported in Table 1, *Publication Age*, *Journal Subject*, *Patent Issued*, and *MTA Issued*, are equal between the treatment and control groups, which is not surprising in light of our requirement for an exact match in these dimensions across the two groups. For the remaining variables, means for the control and treatment groups are much more similar than prior to matching, although differences in means remain statistically significant for four variables (*Journal Impact Factor*, *Citations in* $t-1$, *Citations in* $t-2$, and *Citations Slope*). Differences in distributions are significant for only two variables, *Citations in* $t-1$ and *Citations Slope*, as shown by the *KS* Test. The magnitude and significance of these differences suggests a relatively balanced sample, but one where further controls are needed to ensure that these differences are not driving the result. The coefficient estimates reported in Section 4 suggest that the differences in observable characteristics do not meaningfully influence our estimated effects.

3.2. Estimation

Once we have generated our treatment and control group sample, we report both the raw citation counts and employ a difference-in-differences approach to account for any remaining unobserved differences between the treated and control observations to estimate the size and direction of the treatment effect. More specifically for the latter, we compare the change in the number of citations to one patented publication following the execution of a license (a treated observation) to the change in number of citations for a comparable publication that lacks a license (a matching control observation). A citation by a scientist to his or her own prior publication does not represent a knowledge flow among researchers, and we therefore exclude papers where at least one of the authors of the citing publication is also an author of the cited publication (i.e., a self-citation). We define our outcome of interest as:

$$\begin{aligned}\text{Treatment Effect}_{(t-1)\rightarrow(t+i)} &= (\text{Citations}_{t+i} - \text{Citations}_{t-1})_{\text{pub}_{w/\text{ License}}} \\ &\quad - (\text{Citations}_{t+i} - \text{Citations}_{t-1})_{\text{pub}_{w/o\text{ License}}}\end{aligned} \tag{1}$$

where i is the number of years after the license, ranging from 1 to 3.

The specification is as follows:

[12] Restricting ourselves to control observations that satisfy our matching criteria decreases our sample size. The effects of a smaller sample size on the precision of our estimates, however, are ambiguous. Smaller samples will tend to reduce our precision (making the standard errors larger), but this loss of precision may be offset by greater homogeneity within the smaller sample of treatment and control observations.

[13] Available from the Comprehensive R Archive network at http://CRAN.R-project.org/package=matching (R Core Team, 2014).

[14] Treatment and control variables are in "covariate balance" when they have the same joint distribution of their observed covariates (Diamond and Sekhon, 2013).

[15] Previous studies in the social sciences that utilize genetic matching include Morgan and Harding (2006), Gilligan and Sergenti (2008), Eggers and Mainmueller (2009), Ladd and Lenz (2009), Gordon (2009), and Hopkins (2010).

[16] Coarsened exact matching is one alternative matching approach. Employing a caliper-based method rather than coarsened exact matching, however, allows us to exclude observations whose observable covariates would make them outliers as well as those which would qualify them as "inliers," that is, observations that are within the range of the data, but nevertheless lack a comparable control observation (Sekhon, 2011).

[17] Previous work has shown that patented research differs in quality and type from non-patented research, and it is plausible that citations to papers linked to licensed and non-licensed research disclosures likewise might differ systematically. It is precisely to minimize such otherwise unobserved differences in such citation behavior that we perform the matching procedure. Our matching procedure uses pre-license citations data to ensure that the licensed and non-licensed groups are as similar as we can make them. This procedure makes the pre-license citation trends mechanically equal, but it also imposes a substantive equality: papers linked to both patented and nonpatented disclosures are cited to an equal extent and at the same growth rate. This procedure provides a significantly better "apples-to-apples" comparison than is usually found in similar literature, where linear control terms in Poisson or negative-binomial regressions can only make crude adjustments.

$$\Delta \text{Citations}_{(t-1)\to(t+i)} = \beta_0 + \gamma \text{License} + \beta_1 \text{Age} + \beta_2 \text{Publication Year} + \beta_3 \text{Journal Subject} + \beta_4 \text{Journal Impact Factor} + \beta_5 \text{Citations}_{t-2} + \beta_6 \text{Citations}_{t-1} + \beta_7 \text{MTA Issued} + \beta_8 \text{Patent Granted} + \epsilon \quad (2)$$

Our coefficient of interest is γ, which we present in our results.[18]

Using a difference-in-differences estimator allows us to avoid bias associated with changes that affect the "before" and "after" periods equally. For example, in the matched sample case, if a particular publication receives on average five additional citations per year due to some unobserved covariate, these citations will be included in both $Citations_{t-1}$ and $Citations_{t+1}$ terms, and the impact on the estimate will be zero.

To summarize, our analysis employs two techniques. First, we identify the nearest neighbor to each of our treatment observations using genetic matching. This non-parametric matching technique means that we make fewer assumptions about the parametric form of the effect than we would with a linear or generalized linear specification (e.g., negative binomial). We then estimate the size and direction of the treatment effect by employing a difference-in-differences approach that accounts for any remaining unobserved differences between the treated and control observations that are constant over time. Based on this sample construction, our results should be interpreted as an Average Treatment Effect on the Treated (ATT). In other words, our sample construction implies that our estimates of the effects of licensing on citations have the most external validity for patent-linked publications whose covariates resemble those that receive licenses.

4. Results

4.1. The effects of licensing on citations to patent-linked publications

We now turn to estimating the effects of licensing on scientific communication as measured by citations to patent-linked publications. We first report the findings based on our full sample (588 observations). The effect can be seen directly through the citation pattern for the treatment group (dashed line with circles) and the control group (dotted line with squares), as shown in Fig. 1.

As depicted in the figure, patent-linked publications that are licensed receive more citations immediately after a license is executed than do publications that are linked to unlicensed patents. This difference in the numbers of citations between the treated group and control group slowly increases in years following the license.

Fig. 2 reports results from the difference-in-differences analysis (Eq. (2)), confirming the post-licensing citation pattern presented in Fig. 1 and testing for the statistical significance of the differences in citations between the two groups. The effect of licensing on the number of citations is not statistically significant in the year of the execution of the license (γ = 0.91 with a standard error of 0.57). On average, papers linked to a licensed patent receive 1.46 more annual citations than papers linked to non-licensed patents by the first year (standard error of 1.14), a difference that grows to 2.56 citations by year $t + 2$ (standard error of 1.15) and 3.43 citations by year $t + 3$ (standard error of 2.17). Only the coefficient in year $t + 2$ is barely significant at a 10% level.

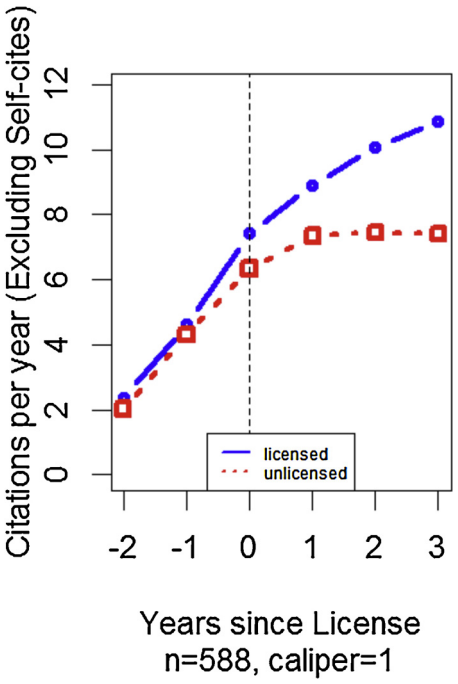


Fig. 1. License effect-full sample.

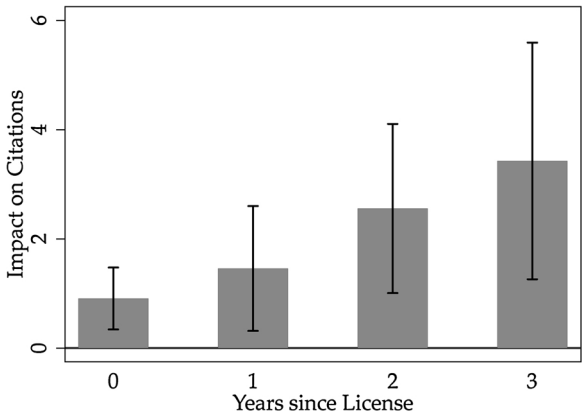


Fig. 2. Regression results of license effect-full sample. (Error bars signify ± 1 standard error.)

This pattern is very similar to the citation trends depicted in Fig. 1, a result that we interpret as reflecting the good covariate balance prior to the difference-in-differences regression.[19]

Inspection of the residuals from our regressions (not reported) shows that the mean and median are aligned and that the residuals are distributed approximately normally, supporting the validity of our model.

The results of this initial analysis of our data suggest that that the negotiation of a license covering a patented academic invention has a modestly positive but not statistically significant effect on follow-on citations by other scientists in subsequent published papers. But this initial analysis does not consider whether the licensed published discovery is a "research tool." In the next section, therefore, we undertake further analysis below by disaggregating our sample into two subsets,

[18] Because we expect the differences in covariates between the treated and control groups after the nearest neighbor matching to be smaller, the linearity assumption embedded in least-squares is more plausible than it would be for an unmatched sample. This assumption nonetheless introduces a lack of flexibility into the specification. Rubin (1979) discusses the value of using these (slightly modified) techniques and concludes that "pair-matching coupled with regression adjustment on the matched pairs is a quite effective general plan for controlling the bias due to matching variables, and this combination is clearly superior to regression adjustment" (p. 318).

[19] We also tested for the overall effect of licensing on citations to related articles by comparing the total number of citations in years 1–3 between articles linked to licensed patents and articles linked to unlicensed patents (unlike our difference-in-differences analysis, this comparison does not control for pre-license citations). Similarly to our results when considering each year separately, we find a positive but not significant result for this comparison (coefficient of +6.26 with a standard error of 4.80).

Table 1
Covariate balance for the full licensing sample.

Covariate	Full licensing sample			
	Mean treated	Mean control	*t*-test	*KS* test
Publication Age	1.88	1.88	–	–
Journal Subject[a]	4.91	4.91	–	–
Patent Issued	0.54	0.54	–	–
MTA Issued	0.04	0.04	–	–
Journal Impact Factor	7.81	7.54	**	–
Publication Year	2000.81	2000.76	–	–
Citations in $t-1$	5.61	4.82	***	***
Citations in $t-2$	2.83	2.33	***	–
Citation slope from $t-2$ to $t-1$	2.78	2.49	**	***

Notes:
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.
[a] *Journal Subject* is a categorical variable, with each subject mapped to a random integer. Therefore the 4.91 values have no literal meaning, but the equality between treatment and control reflects the success of the exact matching.

(1) discoveries that are research tools, which we define as patented disclosures linked to MTAs; and (2) discoveries that are not research tools based on the absence of an MTA.

4.2. Licensing and citations to patented publications associated with research tools

We now focus our attention on the relationship between licensing and citations to publications linked to patented disclosures that we believe are associated with research tools. In their 2007 paper, Murray and Stern, Murray and Stern defined such publications as those linked to any patents within their sample in the 3-digit patent classes 435 (Chemistry: Molecular Biology and Microbiology) and 800 (Multicellular Living Organisms and Unmodified Parts Thereof and Related Processes) as pertaining to research tools. A patent class-based definition, however, does not account for the patent's use. Instead of employing patent classes, as we noted above, we believe that patented publications for which MTAs are issued are more likely to fall within a rough definition of "research tool."

Our analysis of the effects of licensing on this class of publications linked to patented disclosures restricts the sample to only those with material transfer agreements.[20] Thus, both treatment and the control observations have MTAs and the difference between them is whether a license is issued.[21] Table 2 presents the covariate balance for this sample.

As in the full sample, means and distributions for first four variables reported in Table 2, *Publication Age*, *Journal Subject*, *Patent Issued*, and *MTA Issued*, are equal between the treatment and control groups, reflecting our requirement of an exact match. For the remaining variables, only *Journal Impact Factor* and *Citations in* $t-1$ exhibits significant differences in the *t*-test and only *Journal Impact Factor* exhibits a significant difference in the distributions (*KS*) test. We once again interpret the results of this *KS* test as an indication that our matching

[20] Our control sample includes published disclosures with MTAs that are issued before or after the license. We also restricted the sample to instances where an MTA exists prior to the license, so as to ensure against possible reverse causality, and obtained results that are similar in direction, although with much smaller sample sizes. Thus we report results with the larger sample of MTA-linked inventions. This produces a sample of 61 observations.

[21] Because we also match treatment to control observations based on whether the observation has an MTA, treated observations with MTAs prior to license are matched to controls for which an MTA is observed at approximately the same time or earlier, while those that have not received an MTA prior to license are matched to controls that also have not yet received an MTA as of the license execution date.

Table 2
Covariate balance for the research tools sample.

Covariate	Research tools sample			
	Mean treated	Mean control	*t*-test	*KS* Test
Publication Age	1.65	1.65	–	–
Journal Subject	4.65	4.65	–	–
Patent Issued	0.73	0.73	–	–
MTA Issued	0.30	0.30	–	–
Journal Impact Factor	5.45	5.40	***	**
Publication Year	2001.58	2001.28	–	–
Citations in $t-1$	1.85	2.33	–	–
Citations in $t-2$	0.40	0.90	**	–
Citation slope from $t-2$ to $t-1$	1.45	1.43	–	–

Note:
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

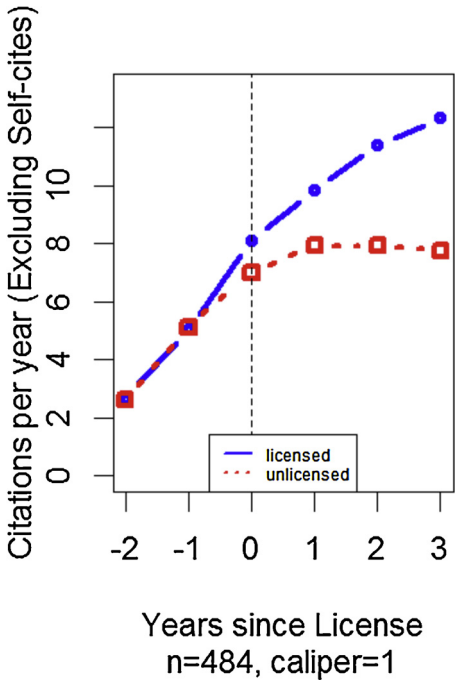


Fig. 3. License effect—non-research tools sample.

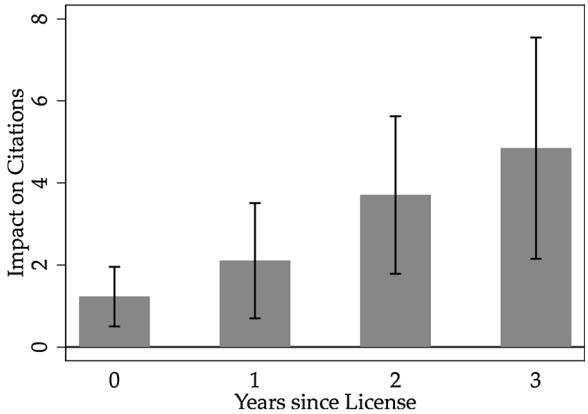


Fig. 4. Regression results of license effect—non-research tools. (Error bars signify ± 1 standard error.)

procedure has produced a relatively balanced sample, but nevertheless include additional controls in our estimates to ensure that these small differences do not drive our results.

Before we report our analysis of citations to "research tool-related" publications, we first report the unadjusted citation curve and regression adjusted results for the subset of observations *not* associated with MTAs. These results are based on 484 observations and are shown in Figs. 3 and 4.[22]

The unadjusted citation curve for the subset of inventions not associated with an MTA (the non-research tools subsample) in Fig. 3 is similar to the citation curve from the overall sample (Fig. 1). As in the overall sample, the citations to these licensed patent-linked publications are higher at the time of the license, and this difference is increasing in the three years after the license. Unlike the results of our empirical analysis for the overall sample, however, these differences are slightly more statistically significant in three of the four time periods covered by our analysis: the coefficient 1.23 is significant at the 9% level at the time of the license, the coefficient of 3.71 is significant at the 5% level in year $t + 2$, and the coefficient 4.85 is significant at the 7% level in year $t + 3$ after the execution of the license. Only the coefficient in year $t + 1$ (2.11) is not significant for the sample of inventions not associated with an MTA.[23]

We now present our analysis of citations to "research tool-related" publications. We report both the unadjusted citation curve for the matched samples in Fig. 5 and the regression adjusted results in Fig. 6 below.

Our difference-in-differences analysis of the effects of licensing on citations to patent-paper pairs that are associated with MTAs reveals a pattern that differs from that observed in the overall sample and the non-research tools subsample. Our earlier results for the overall sample indicated little relationship between licensing and citations to the publications linked to the underlying patented disclosure. But when we disaggregate the sample into MTA-linked and non-MTA-linked discoveries, we find very different results.

For the non-MTA-linked subsample, licensing has a positive, weakly significant correlation with subsequent journal citations. For the MTA-linked sample, however, licensing is associated with a significant decline in citations. Using the covariate adjusted values (Fig. 6), the magnitude of the licensing effect is -1.02 citations in the year of the license (standard error of 0.33), -1.97 citations in year $t + 1$ (standard error of 0.39), -6.57 citations in year $t + 2$ (standard error of 1.07), and -6.58 in year $t + 3$ (standard error of 1.14). The differences in all these years are statistically significant at the 1% level.[24] Overall, these coefficients represent a decrease in citations of more than 50% for the average publication, although our small sample size calls for caution in interpreting the exact magnitude of the effect.[25]

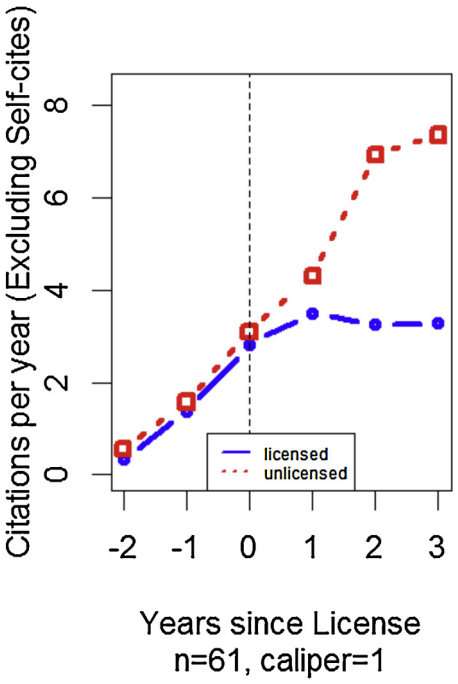


Fig. 5. License effect—research tools.

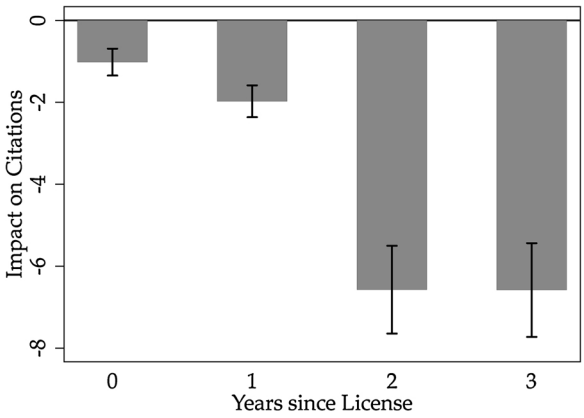


Fig. 6. Regression results of license effect—research tools. (Error bars signify ± 1 standard error.)

5. Discussion and conclusion

This paper has investigated the relationship between licensing and communication among scientists of published research results that are linked to patented academic invention disclosures within the University of California system. Our results suggest that in general, licenses covering patented research disclosures have little impact on the number of citations to related publications (changes in the numbers of citations are not significant for any years in the overall treated sample relative to the control group and only weakly significant (at the 9% confidence level) for a single year in the overall sample). When we disaggregate our sample into those discoveries that are inputs to further research (research tools, i.e., patented disclosures for which MTAs are negotiated) and those discoveries not identified as research tools, however, we find contrasting results. Licensing is associated with a weak but positive effect on follow-on journal citations to articles related to patented disclosures that are not defined as research tools. But licensing has a significantly negative correlation with citations to scientific papers linked to a class of patented invention disclosures that we believe includes a high share of research tools. For these inventions we observe

[22] We also checked the covariate balance between the treated and control groups for the subset of non-MTA-related observations (not reported). As was true of our results for the full sample, there are small differences in the variable means that we address through a differences-in-differences specification.

[23] Similarly to our test for the overall sample, we compared the total number of citations through years 1–3 after the year of the license to articles related to licensed patents with the number of citations to articles related to unlicensed patents for the subset of non-MTA-linked patents. The resulting coefficient (+9.58 with a standard error of 5.86) is just outside the 10% significant level).

[24] As with the overall and non-MTA-linked samples, we compared the total number of citations through years 1–3 after the year of the license to articles related to licensed patents with the number of citations to articles related to unlicensed patents for the subset of MTA-linked patents. The coefficient for licensing (−14.20) is strongly negative and significant (1% level), paralleling our year-by-year results.

[25] We examined whether differences in the make-up of exclusive licenses between research tool (MTA-linked) licensed patents and non-MTA-linked licensed patents could explain the difference in results between the two subsamples by conducting a difference in proportions test for the percentages of exclusive licenses. Eighty-six percent of the licenses of MTA-linked patents are exclusive and 94% of the licenses to non-research tool patents are exclusive. This difference is weakly significant at a 6% level. The similarity in fractions of licenses that are exclusive and the lack of strong statistical significance in this test suggests that a difference between the two samples in the shares of exclusive licenses is not driving our results.

a decrease in the number of citations following a license.

Our results on the effect of licensing on scientific communication expand earlier findings in the literature that examine the effect of licensing on the citation behavior of nonlicensees. Drivas et al. (2017) and Sampat and Ziedonis (2004) found that licensing is associated with increased inventive effort in related fields, assuming that increased citations to the focal patents reflect such inventive effort. We find a similarly positive but nonsignificant effect (at the 5% level) of licensing on citations to papers related to the licensed patent for non-research-tool related discoveries.

Our findings regarding the links between licenses and citations to patented disclosures covering research tools, however, are consistent with other arguments and some evidence that academic licensing of intellectual property may limit dissemination and exploitation of research advances that are important for follow-on research (Heller and Eisenberg, 1998; Walsh et al., 2007). The results of Walsh et al. (2007) underscore the potential impediments to research resulting from denials of access to research materials.

Other anecdotal evidence also suggests that the commercial exploitation of university research advances may be associated with restrictions on access to research materials. One example of the consequences of such a restriction was reported in *Science* (Cohen, 1997). Scientists at Stanford University developed a machine to automate the low-cost synthesis of oligonucleotides, which are critical building blocks of DNA. Prior to the development of this technique, researchers using "oligos" as research inputs had to purchase them from commercial firms at high cost. The availability of low-cost research materials had the potential to accelerate progress in research requiring such inputs, such as in the Human Genome Project. Stanford licensed its technology exclusively to a private firm that sold oligos to researchers but that firm refused to further make the technology available to researchers for their own use. Although the project was completed ahead of its original schedule, genomics researchers argued at the time that this license was "counterproductive to the goal of completing the Human Genome Project as quickly as possible" according to Cohen (p. 1489).

In a more recent episode, scientists studying Rett syndrome, a debilitating disease primarily affecting young girls, have complained about the inability to obtain access to a genetically modified mouse that exhibits traits useful in research on this disease (Hayden, 2011). They blame provisions within the original license that restricted sharing of this input with other researchers. According to Hayden, "scientists and experts in intellectual property issues say that the case shows how science can be impeded when onerous licensing rules govern the sharing of research materials" (p. 318).

How might the licensing of a research tool cause a decline in scientific communication or restrict follow-on research in scientific areas that are reliant on access to such tools? As we note in Section 2.2, researchers who receive requests for materials are more likely to refuse requests when they are involved in related commercial activity, which could include licensing, as Walsh et al. (2007) reported. Such activity could center on the commercialization of a licensed invention linked to a research tool. Campbell et al. (2002) found a similar result in their survey of geneticists-those engaged in "commercial activities" were 72% more likely to deny others' requests for research inputs than scientists not engaged in such activities.

Restriction of scientific communication as a result of licensing may also be an indirect outcome of scientific competition. For example, Strandburg (2008) quotes the Harvard University scientist who developed the Oncomouse as characterizing the terms of the license with DuPont as a "perfect alibi for refusing to send [the mouse] to others—he was no longer a strategic opportunist but rather a scientist beleaguered by legal constraints" (p. 64).

Our data unfortunately do not enable us to discern among these and other possible explanations as underpinning a negative effect of licensing of research tools on subsequent scientific communication and follow-on discovery that depends on access to such materials. This question remains an object for future investigation.

Despite our best efforts to construct comparable treatment and control groups, could uncaptured differences in "quality" between licensed and unlicensed inventions account for our findings? We believe that the contrasting results between the MTA-linked and non-MTA-linked samples should mitigate such a concern. If our licensed publications sample exhibited higher quality than unlicensed publications, we should find higher citations everywhere, which would work against our negative finding in the MTA-linked sample.

Nevertheless, our findings should be interpreted with caution. Our sample of patented publications represents discoveries made by scientists at a single institution, albeit one with nine campuses and five academic medical centers.[26] The University of California has been assigned more U.S. patents than any other U.S. university, and therefore may not be representative of all academic institutions. Our MTA-linked subset of inventions representing research tools is small (61 observations), also limiting our ability to generalize our findings. At the same time, however, our comprehensive matching process generates treatment and control groups that are much better balanced than larger-sample analyses that assume that licensed and unlicensed research are similar (which they often are not). Thus, while we have a small sample, our internal validity is relatively strong, and we are less susceptible to misleading results that arise from incorrect parametric assumptions.

Although our empirical results suggest that the licensing of research tools may impede scientific communication, we are unable to reach strong conclusions about the effects of licensing on scientific communication for the (potentially) much larger set of research tools not covered by MTAs. Moreover, our sample of MTA-linked disclosures is limited in scope. Developing clearer and empirically tractable measures that enable the identification of research tools is an important task for future research.

This study makes three contributions. First, it introduces a novel method for linking patents and publications associated with an academic research advance, and thereby automates a previously arduous process that has been difficult to scale up. Second, it employs a matching methodology that avoids strong and potentially inappropriate parametric assumptions about the complex interactions that underpin journal citation patterns. And finally, but most importantly, this paper contributes to our understanding of the effect of the licensing of intellectual property rights covering academic research output on scientific communication.

Supplementary data and analysis.

Supplementary data and analysis associated with this article can be found, in the online version, at https://doi.org/10.1016/j.respol.2018.03.008.

[26] Moreover, the UC system is widely acknowledged to be one of the leading research performers in the United States and arguably, the global economy. Five UC campuses, UC Berkeley, UC San Francisco, UC San Diego, UCLA, and UC Santa Barbara) were included in the 2017 listing of the world's top fifty universities compiled by Shanghai Jiaotong University (http://www.shanghairanking.com/ARWU2017.html; consulted 12/14/17), and the UC system received more fiscal 2016 federal research funding through the National Science Foundation and the National Institutes of Health than any other U.S. university (http://www.ucop.edu/federal-government-relations/_files/Advocacy/Federal-Research/Fact_Sheet_Federal_Investment_in_Research_Final.pdf; consulted 12/14/17). Although our empirical results thus cover a single public university system, it is one that is a leader in research performance, and results for this system may well apply to other leading U.S. research universities.

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